



SUSTAINABLE OLYMPIAD

Sustainable Olympiad 2024

Round 1: Online Knowledge Challenge

Junior Division (ages 12-14)

Time allowed	90 minutes
Total marks	50
Part A	10 multiple-choice questions, 20 marks
Part B	4 structured problems, 30 marks

Instructions

- Answer **all** questions. Work together as a team.
- Part A: choose **one** answer (A, B, C or D) for each question. There are no negative marks.
- Part B: show all your working. Marks are given for correct method even if the final answer is wrong.
- Give numerical answers to an appropriate number of significant figures, with units.
- Scientific calculators are allowed. Internet access and AI tools are not allowed during the round.
- A data sheet is provided on the next page.

Team name

School

Data sheet

- 1 kWh = 1000 Wh = 3.6 MJ; 1 tonne (t) = 1000 kg
- 1 litre of water has a mass of 1 kg
- Heating 1 kg of water by 1 °C needs 4.18 kJ
- 1 hectare (ha) = 10 000 m²; 1 km² = 100 ha

Part A: Multiple choice (20 marks)

Each question is worth 2 marks. Choose one answer.

- A1.** A 2000 W kettle is switched on for 3 minutes. How much energy does it use? [2]
A 0.01 kWh
B 0.1 kWh
C 6 kWh
D 360 kWh
- A2.** In which layer of the atmosphere is most of the ozone layer found? [2]
A Troposphere
B Stratosphere
C Mesosphere
D Thermosphere
- A3.** Which of these crops does NOT depend on insects for pollination? [2]
A Apple
B Almond
C Wheat
D Strawberry
- A4.** A plant has plenty of water in its soil. What happens to its rate of transpiration on a hot, dry, windy day? [2]
A It decreases, because wind closes the stomata
B It increases, because water vapour diffuses away from the leaves faster
C It stays the same
D It stops completely
- A5.** Last year a town produced 600 t of waste and recycled 30% of it. This year the waste fell by 10% and the recycling rate rose to 40%. How much more waste was recycled this year? [2]
A 18 t
B 36 t
C 54 t
D 60 t

- A6.** Making one cotton T-shirt uses about 2700 L of water. A class of 30 students each buys 2 fewer T-shirts this year. How much water is saved? **[2]**
- A** 5400 L
 - B** 81 000 L
 - C** 162 000 L
 - D** 1 620 000 L
- A7.** What causes coral bleaching? **[2]**
- A** Corals expel the algae living in their tissues when the water becomes too warm
 - B** Strong sunlight bleaches the coral's pigment
 - C** Chlorine pollution from swimming pools
 - D** Fish eat the colourful algae on the reef
- A8.** Which item takes the longest to break down in the sea? **[2]**
- A** Paper bag
 - B** Aluminium can
 - C** Plastic bottle
 - D** Glass bottle
- A9.** On a 1:50 000 map, a forest is a rectangle 3 cm by 2 cm. What is its real area? **[2]**
- A** 6 ha
 - B** 15 ha
 - C** 150 ha
 - D** 1500 ha
- A10.** Which shows the energy transfers in a hydroelectric power station? **[2]**
- A** Gravitational potential → kinetic → electrical
 - B** Chemical → thermal → electrical
 - C** Kinetic → gravitational potential → electrical
 - D** Thermal → kinetic → electrical

Part B: Structured problems (30 marks)

B1. A shorter shower (8 marks)

Leo takes an 8-minute shower every day with a shower head that uses 12 L per minute. He switches to a 5-minute shower with an eco shower head that uses 8 L per minute. The water is heated from 15 °C to 40 °C.

(a) How much water does Leo save each day? **[2]**

(b) How much water does he save in a year? **[2]**

(c) Calculate the energy saved in a year by not heating this water, in kWh. **[4]**

B2. Can trees cancel our emissions? (7 marks)

A growing tree absorbs on average 22 kg of CO₂ per year. A family's car and heating emit 7.5 t of CO₂ per year. A forest has about 400 trees per hectare.

(a) How many trees are needed to absorb the family's emissions? **[2]**

(b) What area of forest is this, in hectares? **[2]**

(c) Give two reasons why planting trees alone is not a solution to climate change. **[3]**

B3. Counting daisies (7 marks)

A field measures 50 m by 40 m. Students place ten 1 m² quadrats at random and count the daisies in each: 4, 7, 3, 0, 6, 5, 8, 2, 5, 10.

(a) Calculate the area of the field. **[1]**

(b) Calculate the mean number of daisies per m². **[2]**

(c) Estimate the total number of daisies in the field. **[2]**

(d) Why should the quadrats be placed at random, and why is it better to use more quadrats? **[2]**

B4. Electric or petrol? (8 marks)

An electric car uses 16 kWh per 100 km. Electricity costs €0.22 per kWh and causes 0.25 kg CO₂ per kWh. A petrol car uses 6.0 L per 100 km. Petrol costs €1.80 per litre, and burning 1 L releases 2.3 kg CO₂.

(a) Calculate the fuel cost per 100 km for each car. **[2]**

(b) Calculate the CO₂ per 100 km for each car.

[2]

(c) A family drives 12 000 km a year. How much money would they save on fuel with the electric car?

[2]

(d) Give two reasons why the electric car's real climate impact is higher than your answer to (b) suggests.

[2]

END OF PAPER